

Multimodal Fake News Detection: A Review of Machine Learning, Deep Learning, and Multimodal Fusion Approaches

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Abstract—The rapid growth of social media and digital communication platforms has significantly increased the volume and speed at which misleading and fabricated information can be disseminated. Consequently, automated fake news detection has become an important research area involving natural language processing, computer vision, machine learning, and multimodal learning. This paper presents a review of twelve recent studies covering traditional machine learning, deep learning, multimodal feature fusion, attention mechanisms, contrastive learning, large language models, emotional and linguistic feature analysis, and logic-based approaches for fake news detection. The reviewed studies demonstrate the progression from handcrafted textual and linguistic features toward multimodal architectures capable of jointly processing textual and visual information. Particular attention is given to the datasets, feature extraction strategies, fusion mechanisms, classification models, evaluation measures, and reported performance of the reviewed approaches. The review also identifies recurring limitations, including dependence on domain-specific datasets, computational complexity, restricted modality coverage, insufficient interpretability, and difficulties in modelling relationships between different modalities. Finally, the paper summarizes open research challenges and directions for developing more robust, generalizable, and interpretable fake news detection systems.

Index Terms—fake news detection, misinformation, machine learning, deep learning, multimodal learning, multimodal fusion, natural language processing, computer vision

I. INTRODUCTION

The rapid expansion of social media, online news platforms, and digital communication has transformed the way information is created, distributed, and consumed. Although these technologies provide efficient access to information, they also enable misleading or fabricated information to spread rapidly. Fake news can influence public opinion, create confusion, damage reputations, and produce social consequences. Consequently, automated fake news detection has become an important research problem.

Early fake news detection approaches primarily concentrated on textual content and handcrafted linguistic features. Traditional machine learning algorithms such as Support Vector Machines, Random Forests, Logistic Regression, Decision

Trees, and Naive Bayes were commonly used for classification. More recent research has introduced deep learning models capable of automatically learning semantic representations from news content.

The increasing presence of images and other forms of multimedia in online news has further motivated multimodal fake news detection. Multimodal approaches attempt to exploit complementary information from text and visual content rather than relying on a single modality. Recent studies have therefore investigated attention mechanisms, cross-modal semantic alignment, contrastive learning, image captioning, feature fusion, and large language models.

This paper reviews twelve studies representing different directions within fake news detection. The reviewed works include traditional machine learning approaches based on linguistic and emotional features, multimodal fusion architectures, semantic alignment methods, LLM-enhanced detection, and logic-based approaches.

The remainder of this paper is organized as follows. Section II discusses the development of fake news detection approaches. Section III reviews the selected studies. Section IV provides a comparative analysis. Section V discusses major research challenges. Section VI presents future research directions, and Section VII concludes the review.

II. BACKGROUND AND EVOLUTION OF FAKE NEWS DETECTION

Fake news detection research has evolved from manually designed features toward representation learning and multimodal reasoning. Traditional approaches generally rely on observable characteristics of textual content, such as word frequencies, n-grams, linguistic patterns, sentiment, and stylistic characteristics.

Balakrishnan et al. investigated textual and linguistic features including word bigrams and part-of-speech information and evaluated several traditional machine learning classifiers. Their findings demonstrated that textual and linguistic features

can contribute effectively to fake news detection, while combining different feature groups does not necessarily provide a significant improvement [1].

Similarly, emotion-oriented analysis has been explored as an additional source of information. Farhoudinia et al. studied sentiment and emotional characteristics of COVID-19-related fake news and reported that fake news tended to be associated with more negative emotional characteristics. Machine learning models were evaluated with and without emotional features to investigate their contribution to detection performance [2].

The development of deep learning introduced automatic representation learning, reducing dependence on manually engineered features. Transformer-based architectures, convolutional networks, recurrent networks, and attention mechanisms have subsequently become important components of fake news detection systems.

A systematic mapping study by Choras et al. reviewed the broader application of machine learning to online disinformation detection. The study highlighted the use of multiple information sources, including text, images, network information, and source or author characteristics [3].

The evolution toward multimodal learning is particularly important because online news frequently combines text and images. Multimodal methods attempt to capture both modality-specific information and relationships between modalities.

III. REVIEW OF SELECTED STUDIES

A. Multi-Level Fusion-Based Framework

Abbas and Taeihagh proposed a multi-level fusion-based framework for multimodal fake news classification. The framework combines convolutional neural networks with dual convolution layers and recurrent neural networks to extract visual and textual features. Different fusion strategies, including mean, weighted-mean, maximum, and sum fusion, were investigated. A polynomial-kernel classification module was subsequently used to classify news as fake or real [4].

The study evaluated the framework using five datasets, including ISOT, Fake vs. Real News, WELFake, FA-KES, and Twitter. The reported accuracy values were 97.25%, 91.07%, 98.16%, 54.03%, and 91.63%, respectively. The results demonstrate the potential of combining textual and visual representations through multiple fusion mechanisms.

B. Systematic Mapping of Machine Learning Approaches

Choras et al. presented a systematic mapping study of machine learning techniques for fake news and online disinformation detection [3]. Rather than introducing a single classification architecture, the study analyzed the existing body of research, datasets, methodological directions, and challenges.

The review showed that fake news detection can involve several types of digital information, including textual content, images, network information, and source-related characteristics. The study also identified methodological gaps and emphasized the need for further research into reliable machine learning-based detection methods.

C. Intra-Modality Feature Aggregation and Inter-Modality Semantic Fusion

Zhu et al. proposed the IFIS framework, which addresses both intra-modality feature aggregation and inter-modality semantic fusion [5]. The method uses entity detection and attention mechanisms to aggregate features within individual modalities. Inter-modality semantic fusion is performed using two parallel co-attention blocks.

The method was evaluated on the Weibo and Twitter datasets. The authors reported that the proposed approach achieved improved performance compared with several existing multimodal fake news detection methods. The study demonstrates the importance of modelling both within-modality information and relationships between modalities.

D. Stylometric and Semantic Similarity Oriented Detection

Nadeem et al. proposed the Stylometric and Semantic similarity oriented Multimodal fake news detection (SSM) framework [6]. The architecture consists of multiple modules covering textual stylometric feature learning, headline and news summary similarity, image semantic similarity, image forgery detection, and feature fusion and classification.

The method employs a Hyperbolic Hierarchical Attention Network to capture textual style and content patterns. For visual analysis, the framework extracts semantic information and incorporates image forgery analysis using Error Level Analysis and a pretrained EfficientNetB7 model. The resulting textual and visual information is combined for fake news classification.

E. SARD: CLIP Contrastive Learning and Semantic Alignment

Yan et al. introduced SARD, a fake news detection method based on CLIP contrastive learning and multimodal semantic alignment [7]. The framework combines textual and visual representations with social-context information represented through a news-oriented heterogeneous information network.

The method performs cross-modal alignment between textual and visual features and additionally aligns news content with social-context features. Evaluation on three real-world datasets demonstrated improvements over twelve state-of-the-art competitors, with reported average improvements of 2.13 percentage points in accuracy and 2.89 percentage points in Macro-F1 compared with leading baseline models.

F. Bridging the Semantic Gap Between Modalities

Liu et al. proposed a multimodal fake news detection method designed to reduce the semantic gap between text and images [8]. The approach generates image captions and incorporates the resulting image descriptions into the textual representation. Global image features and object-level visual features are also combined to improve image utilization.

A multimodal Transformer is subsequently used to integrate the resulting representations. Experiments were conducted on two publicly available datasets, and the authors reported that the proposed method improved performance compared with existing methods.

G. Emotion-Aware COVID-19 Fake News Detection

Farhoudinia et al. investigated whether sentiment and emotional features can improve the detection of COVID-19 fake news on social media [2]. The study compared the emotional characteristics of real and fake news using a labelled COVID-19 Twitter dataset.

Sentiment categories and eight basic emotions were extracted using different sentiment and emotion lexicons. Random Forest, Support Vector Machine, and Naive Bayes models were evaluated with and without emotional features. The study reported that incorporating emotional information improved detection performance.

H. COVID-19 Health Information Classification

Sharifpoor et al. investigated machine learning and deep learning methods for classifying and fact-checking health-related information on Twitter/X [9]. The study compared several classification approaches, including Decision Tree, Random Forest, Support Vector Machine, GRU, LSTM, TextConvoNet, and a stacking ensemble.

The reported results indicate that TextConvoNet achieved 90.28% accuracy, 90.29% F1-score, and 0.901 ROC-AUC. The work demonstrates the effectiveness of deep learning approaches for domain-specific health misinformation detection while also highlighting the importance of considering the characteristics of the target domain.

I. LLM-Enhanced Multimodal Detection

Wang et al. proposed an LLM-enhanced multimodal fake news detection framework, referred to as FND-LLM [10]. The framework contains specialized branches for textual features, visual semantic features, visual tampering features, cross-modal relationships, and large-language-model-based reasoning.

The method combines BERT and visual models with CLIP, co-attention, and an LLM reasoning branch. The LLM branch provides auxiliary reasoning and explanations, while multimodal representations are further integrated before classification.

Experiments were conducted on Weibo, GossipCop, and Politifact. The authors reported improvements in overall accuracy of 0.7%, 6.8%, and 1.3%, respectively, compared with the corresponding existing approaches.

J. Logic-Based Fake News Diffusion Analysis

Fionda proposed a logic-based approach to fake news detection based on temporal diffusion patterns in social networks [11]. The approach introduces News-Diffusion Temporal Logic (NDTL) and constructs propagation graphs from information diffusion patterns.

Rather than relying primarily on deep neural representations, the method uses temporal logic rules and model checking to identify potentially fake information at an early stage. Experiments were performed using Twitter15 and Twitter16 datasets as well as synthetic diffusion graphs. The reported F1-scores were 0.664 on Twitter15 and 0.613 on Twitter16.

K. Content and Linguistic Feature-Based Detection

Balakrishnan et al. examined the use of textual and linguistic content features for COVID-19 fake news detection [1]. The study considered features such as bigrams, part-of-speech distributions, type-token ratio, punctuation, stop words, sentence length, and uppercase characters.

Decision Tree, K-Nearest Neighbor, Logistic Regression, Support Vector Machine, and Random Forest classifiers were evaluated using a COVID-19 fake news dataset. Ten-fold cross-validation was used for evaluation. The reported results showed that Support Vector Machine achieved the highest accuracy of 97.48% under the reported POS-feature configuration, while Random Forest achieved 96.69%.

L. Multimodal Attention and Residual Convolutional Networks

Lu and Yao proposed a multimodal fake news detection model combining residual convolutional networks with multi-head attention mechanisms [12]. The architecture is designed to process text, image, and video information and includes multimodal feature fusion and cross-modal alignment mechanisms.

Residual connections are used to improve the depth and stability of the network, while multi-head attention is used to emphasize relevant features. According to the reviewed literature, the model achieved 97.7% accuracy. The approach demonstrates the potential of combining attention, residual learning, and multimodal feature fusion.

Table I summarizes the major characteristics of the reviewed studies.

IV. COMPARATIVE ANALYSIS

The reviewed studies demonstrate a clear progression from single-modality machine learning approaches toward increasingly complex multimodal architectures.

Traditional machine learning methods remain useful when the input domain is relatively constrained. Linguistic and emotional features can provide useful information for specialized datasets, as demonstrated by the COVID-19-focused studies.

However, multimodal approaches generally attempt to exploit information that cannot be represented by text alone. Methods such as IFIS, SSM, SARD, and the multimodal Transformer approach explicitly model relationships between textual and visual information.

Recent approaches increasingly emphasize semantic interaction rather than simple feature concatenation. Attention mechanisms, CLIP-based alignment, image captioning, co-attention, and contrastive learning are examples of techniques designed to reduce inconsistencies between modalities.

Large language models represent another emerging direction. The FND-LLM framework demonstrates how LLM reasoning can complement smaller language models and multimodal representations. However, such approaches can introduce substantial computational requirements.

Logic-based approaches provide a different perspective. Instead of depending entirely on neural feature representations,

TABLE I
COMPARISON OF REVIEWED FAKE NEWS DETECTION APPROACHES

Study	Year	Primary Modality	Main Technique	Reported Result
Abbas and Taeihagh	2025	Text + Image	Multi-level CNN-RNN fusion	Up to 98.16% accuracy
Choras et al.	2021	Multiple	Systematic mapping of ML methods	Review study
Zhu et al.	2024	Text + Image	Intra-modal aggregation + co-attention	Improved over competing methods
Nadeem et al.	2023	Text + Image	Stylometry + semantic similarity + forgery detection	Multimodal classification
Yan et al.	2024	Text + Image + Social context	CLIP contrastive learning	+2.13% accuracy over leading baseline
Liu et al.	2023	Text + Image	Image captioning + Transformer	Improved over existing methods
Farhoudinia et al.	2024	Text	Sentiment + emotion features	Improved detection with emotion features
Sharifpoor et al.	2025	Text	ML/DL classifiers	90.28% accuracy
Wang et al.	2024	Text + Image	LLM + CLIP + co-attention	Up to +6.8% accuracy improvement
Fionda	2025	Propagation data	Temporal logic	F1 = 0.664 / 0.613
Balakrishnan et al.	2023	Text	Linguistic features + ML	97.48% accuracy
Lu and Yao	2025	Text + Image + Video	Residual CNN + attention	97.7% accuracy

temporal diffusion models can exploit the propagation structure of information through social networks.

V. RESEARCH CHALLENGES

A. Limited Generalization

Many reviewed studies rely on datasets associated with specific domains, platforms, or events. COVID-19 misinformation is a notable example. Models trained on domain-specific information may not generalize effectively to unrelated topics.

B. Cross-Modal Semantic Inconsistency

Multimodal fake news detection introduces the additional challenge of determining whether text and visual content are semantically consistent. Simple feature concatenation may fail to capture subtle relationships between modalities.

The use of co-attention, contrastive learning, image captioning, and semantic alignment in recent studies demonstrates the importance of this issue.

C. Computational Complexity

Modern multimodal systems may contain several deep neural networks, attention modules, pretrained vision-language models, and large language models. Although these architectures can improve detection performance, their computational requirements can make deployment more difficult.

D. Interpretability

A major challenge remains the interpretation of model predictions. Some approaches provide additional reasoning or

explanation mechanisms, while many multimodal architectures operate largely as black-box classifiers.

Interpretability is particularly important in misinformation detection because incorrect predictions can affect the perceived credibility of information.

E. Modality Coverage

Most reviewed multimodal systems focus primarily on text and images. Only some recent approaches explicitly consider video. Consequently, there remains substantial scope for methods capable of handling different forms of digital information within a unified framework.

F. Dataset Quality

The performance of fake news detection models depends heavily on the quality and representativeness of datasets. Differences in annotation strategies, class distributions, domains, and data sources make direct comparison between reported results difficult.

VI. FUTURE RESEARCH DIRECTIONS

Several directions emerge from the reviewed literature.

First, future research can investigate larger and more diverse datasets covering multiple domains and information sources. This can help improve the generalization of fake news detection models.

Second, multimodal models can move beyond simple feature combination toward deeper semantic interaction. Cross-modal attention, contrastive learning, and alignment mechanisms provide promising directions.

Third, efficient multimodal architectures are needed to reduce the computational cost associated with large pretrained models and multiple feature extraction networks.

Fourth, explainability should become an integral component of fake news detection. Systems should provide evidence or reasoning that allows users to understand why a prediction was produced.

Finally, future research should investigate broader multimedia settings, including video and other content types, while maintaining reliable cross-modal semantic relationships.

VII. CONCLUSION

This paper reviewed twelve studies covering major developments in fake news detection using machine learning, deep learning, and multimodal approaches. The reviewed literature demonstrates a transition from manually engineered linguistic features toward learned representations, multimodal fusion, semantic alignment, attention mechanisms, contrastive learning, and large language model reasoning.

Traditional machine learning remains effective for specialized textual datasets, while multimodal approaches provide additional information by jointly analysing different content types. Recent architectures increasingly focus on modelling relationships between modalities rather than simply processing each modality independently.

Despite substantial progress, important challenges remain. These include domain generalization, dataset quality, computational complexity, cross-modal inconsistencies, limited modality coverage, and interpretability. Addressing these challenges will be important for developing reliable and practical fake news detection technologies.

REFERENCES

- [1] V. Balakrishnan, H. L. Zing, and E. Laporte, "COVID-19 infodemic – Understanding content features in detecting fake news using a machine learning approach," *Malaysian Journal of Computer Science*, vol. 36, no. 1, 2023, doi: 10.22452/mjcs.vol36no1.1.
- [2] B. Farhoudinia, S. Ozturkcan, and N. Kasap, "Emotions unveiled: detecting COVID-19 fake news on social media," *Humanities and Social Sciences Communications*, vol. 11, no. 640, 2024, doi: 10.1057/s41599-024-03083-5.
- [3] M. Choras, K. Demestichas, A. Gielczyk, A. Herrero, P. Ksieniewicz, K. Remoundou, D. Urda, and M. Woźniak, "Advanced machine learning techniques for fake news (online disinformation) detection: A systematic mapping study," *Applied Soft Computing*.
- [4] F. Abbas and A. Taeihagh, "A multi-level fusion-based framework for multimodal fake news classification using semantic feature extraction," *International Journal of Machine Learning and Cybernetics*, vol. 16, pp. 6531–6560, 2025, doi: 10.1007/s13042-025-02633-w.
- [5] P. Zhu, J. Hua, K. Tang, J. Tian, J. Xu, and X. Cui, "Multimodal fake news detection through intra-modality feature aggregation and inter-modality semantic fusion," *Complex & Intelligent Systems*, vol. 10, pp. 5851–5863, 2024, doi: 10.1007/s40747-024-01473-5.
- [6] M. I. Nadeem, K. Ahmed, Z. Zheng, D. Li, M. Assam, Y. Y. Ghadi, F. H. Alghamedy, and E. T. Eldin, "SSM: Stylometric and semantic similarity oriented multimodal fake news detection," *Journal of King Saud University – Computer and Information Sciences*, 2023, doi: 10.1016/j.jksuci.2023.101559.
- [7] F. Yan, M. Zhang, B. Wei, K. Ren, and W. Jiang, "SARD: Fake news detection based on CLIP contrastive learning and multimodal semantic alignment," *Journal of King Saud University – Computer and Information Sciences*, vol. 36, 102160, 2024, doi: 10.1016/j.jksuci.2024.102160.
- [8] P. Liu, W. Qian, D. Xu, B. Ren, and J. Cao, "Multi-modal fake news detection via bridging the gap between modals," *Entropy*, vol. 25, no. 4, p. 614, 2023, doi: 10.3390/e25040614.
- [9] E. Sharifpoor, M. Okhovati, M. Ghazizadeh-Ahsaee, and M. Avaz Beigi, "Classifying and fact-checking health-related information about COVID-19 on Twitter/X using machine learning and deep learning models," *BMC Medical Informatics and Decision Making*, vol. 25, no. 73, 2025, doi: 10.1186/s12911-025-02895-y.
- [10] J. Wang, Z. Zhu, C. Liu, R. Li, and X. Wu, "LLM-enhanced multimodal detection of fake news," *PLOS ONE*, vol. 19, no. 10, e0312240, 2024, doi: 10.1371/journal.pone.0312240.
- [11] V. Fionda, "Logic-based analysis of fake news diffusion on social media," *Social Network Analysis and Mining*, vol. 15, no. 59, 2025, doi: 10.1007/s13278-025-01472-5.
- [12] Y. Lu and N. Yao, "A fake news detection model using the integration of multimodal attention mechanism and residual convolutional network," *Scientific Reports*, vol. 15, no. 20544, 2025, doi: 10.1038/s41598-025-05702-w.